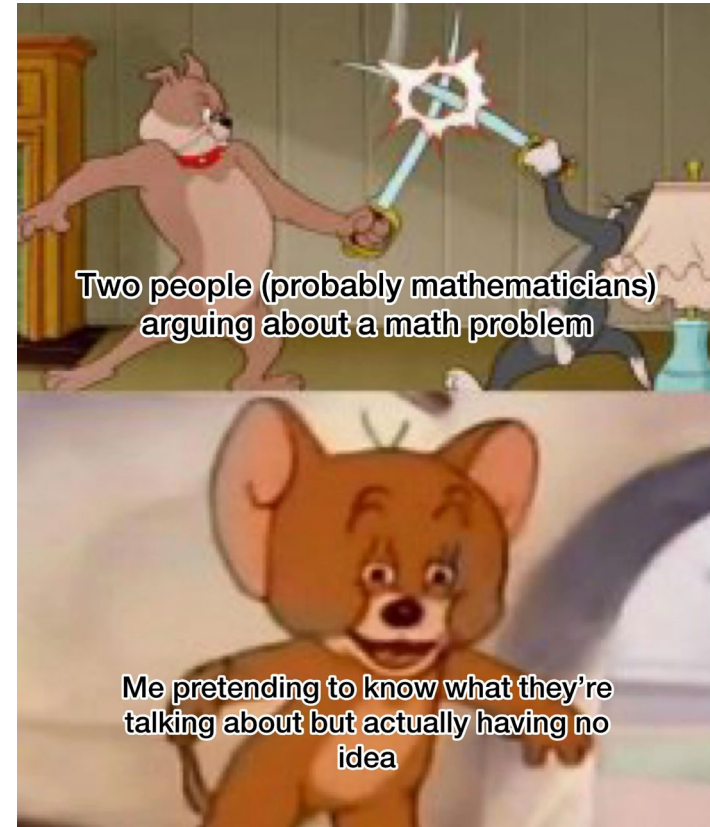


Model Notation

Bollen Notation

- In this class we will use notation introduced by Bollen (2026)
 - Similar to “all y” LISREL notation
- There are alternative and perfectly correct notation systems
 - Many notations are specific to a software
 - Bentler-Weeks notation for EQS (Bentler & Weeks, 1979, 1980, 1982, 1985; Bentler, 1983a,b)
 - RAM (Reticular Action Model) notation (McArdle, 2005)
 - COSAN, COvariance Structural ANalysis (McDonald, 1978)
- For most purposes the notation you use does not make a difference, but there are special cases that some notations allow and others don't.



Elements of an SEM

1. Random Variables

- Variables assumed to differ across cases within a sample
- Three types:
 - a) Latent
 - b) Observed
 - c) Disturbance/Error (note: “residual” is saved for something else)

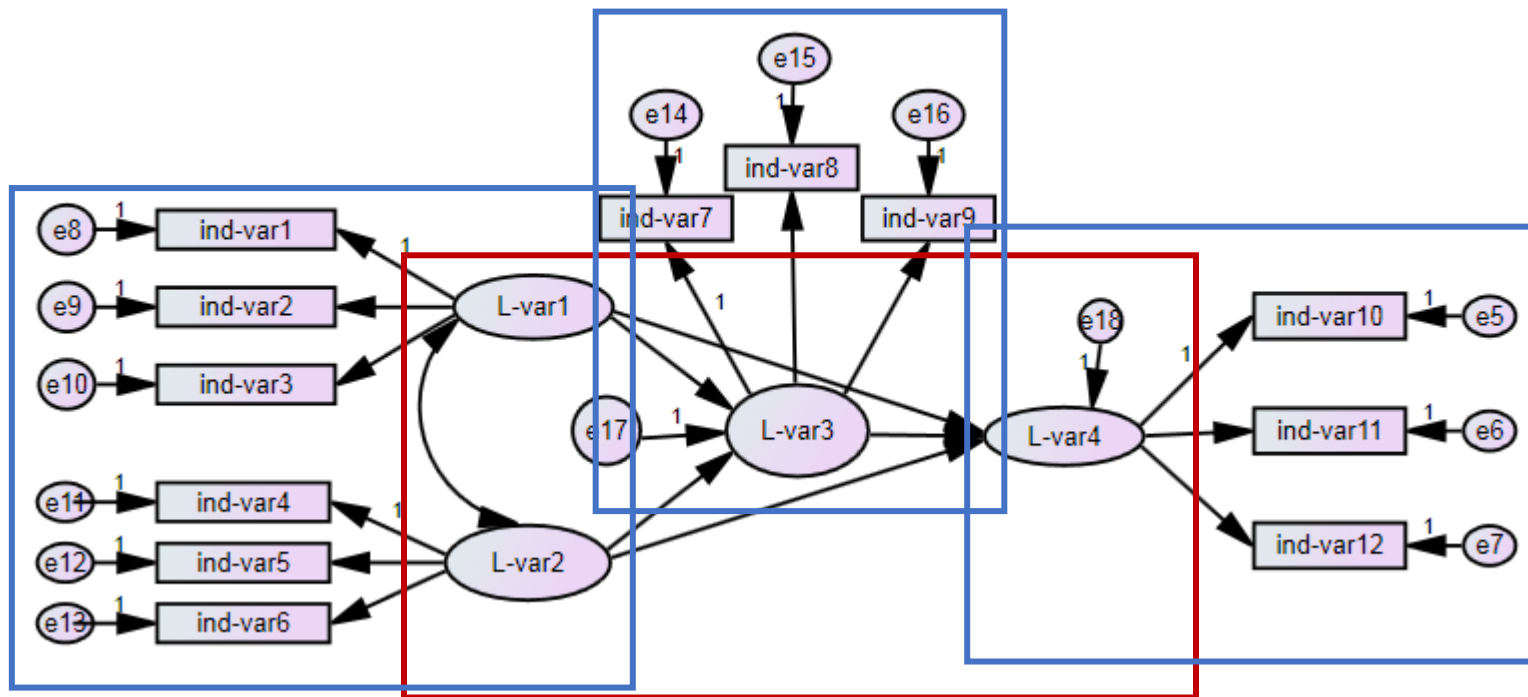
2. Structural parameters

- links between variables
- constant (for models we'll deal with, but see multilevel SEM)
- sometimes described as “causal”

Subsystems of an SEM

There are two major parts of an SEM

1. Latent Variable Model: relations among latent variables
2. Measurement Model: relations among observed and latent variables



What is a latent variable?

A variable which captures variation with **no error** (“purest form”)

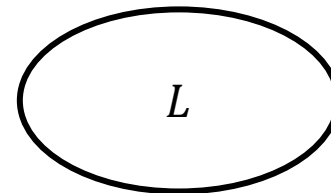
We often want to measure something difficult to observe:

- Depression
- Attention
- Self-regulation

We cannot directly observe this without **measurement error**

Latent Variables are...

- Unidimensional
- Measured without error



Represented in a circle/oval in path diagrams

AKA: Unobserved, unmeasured, **factors** (not the same as “factor” within R)

Latent Variables vary in Abstractness

It's tempting to think that all latent variables are abstract.

Even non-abstract things can be latent if they are measured with error.



Height can be latent

In a study asking men to self-report height, they may over (or under) report their height



Weight can be latent

When weighing rocks for a geology study, the scale is **so sensitive** and it was **so windy**, we weighed each rock 3 times. Still there will be error.

Two Type of Latent Variables

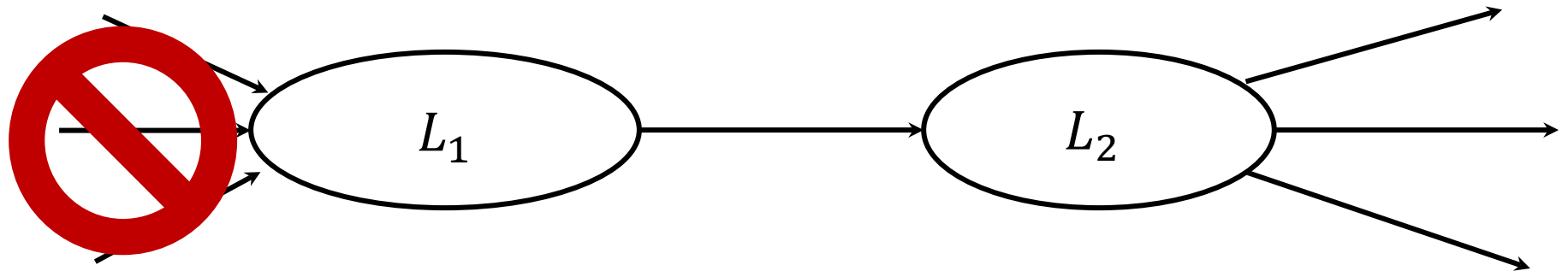
Exogenous

- Exo = outside
- The causes of the variable are *outside* the model
- Only a predictor (right hand side of regression equation), never an outcome (left side)

Endogenous

- Endo = inside
- The causes of the variable are *inside* the model
- Can be a predictor (right side), must be an outcome (left side)

Some notations differentiate, but we will not



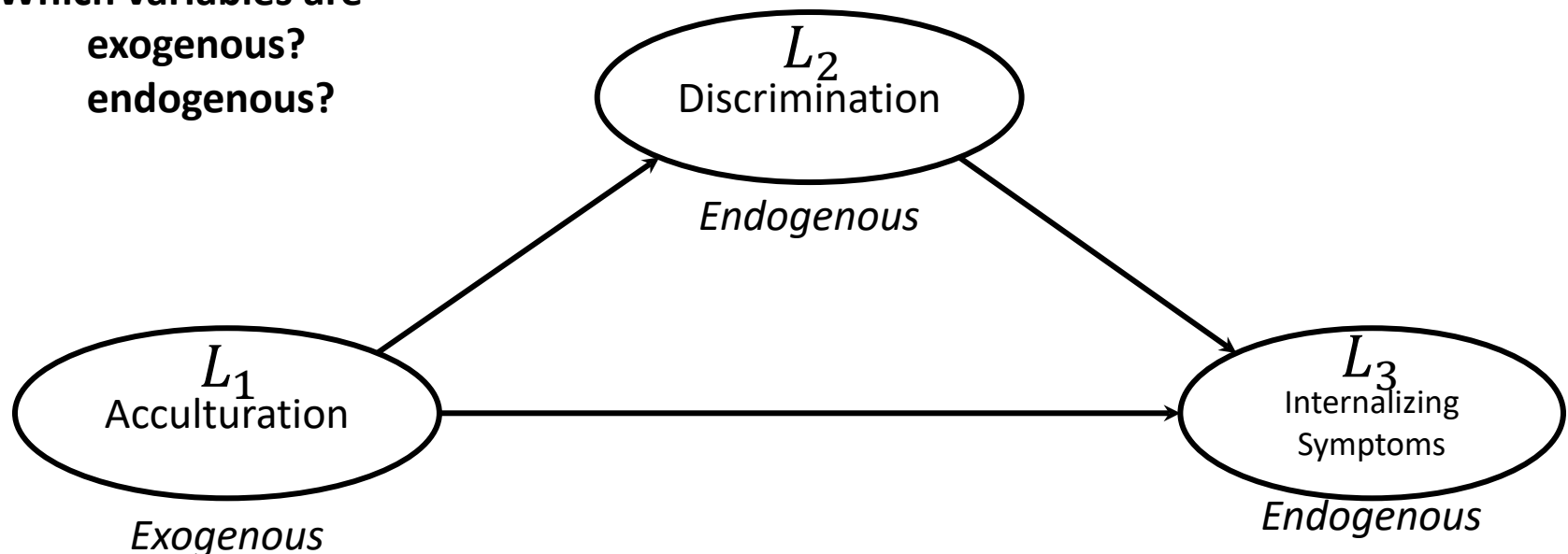
An Example: Internalizing among Latino Youth

Consider the following hypothesis:

- Discrimination predicts internalizing symptoms among latino youth
- Acculturation predicts both discrimination and internalizing

We're not technically doing path diagrams yet... but just to get the picture

Which variables are
exogenous?
endogenous?



A third “type”: unexplained component

In latent variable models “errors”/ “deviations” are represented as latent variables

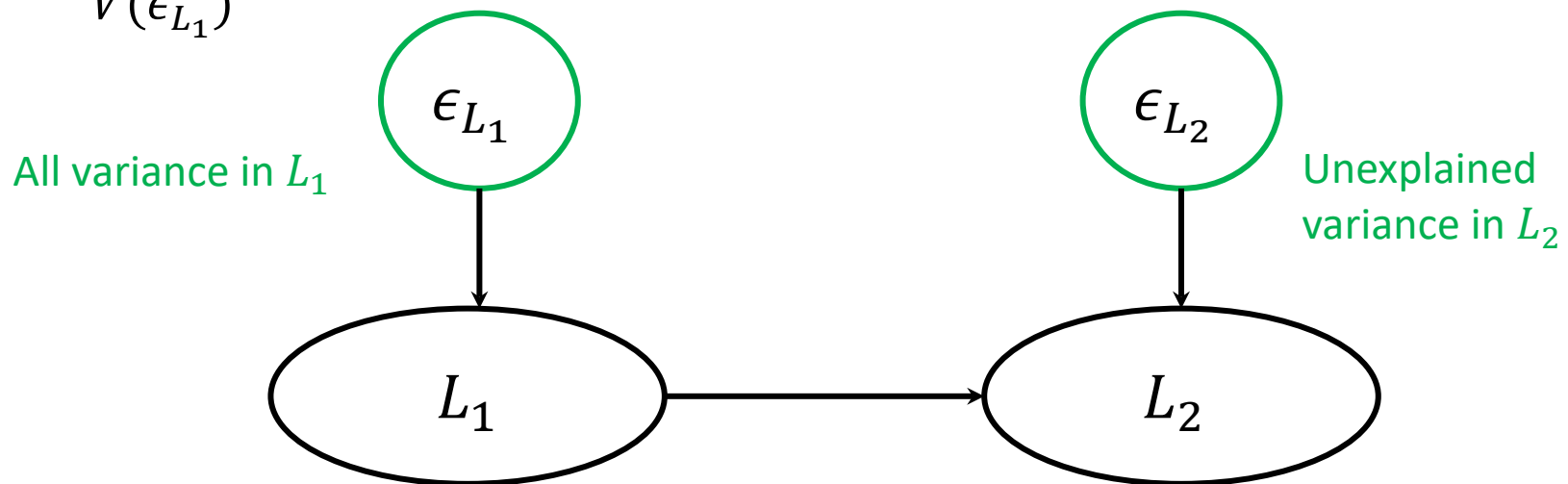
The term “residual” is **not used** for this component (reserved for something else)

Notation: ϵ , epsilon

All variables have error terms

Represents all variables that influence L but are omitted from the model

For exogenous variables, this represents all variability in L_1 (i.e., $V(L_1) = V(\epsilon_{L_1})$)



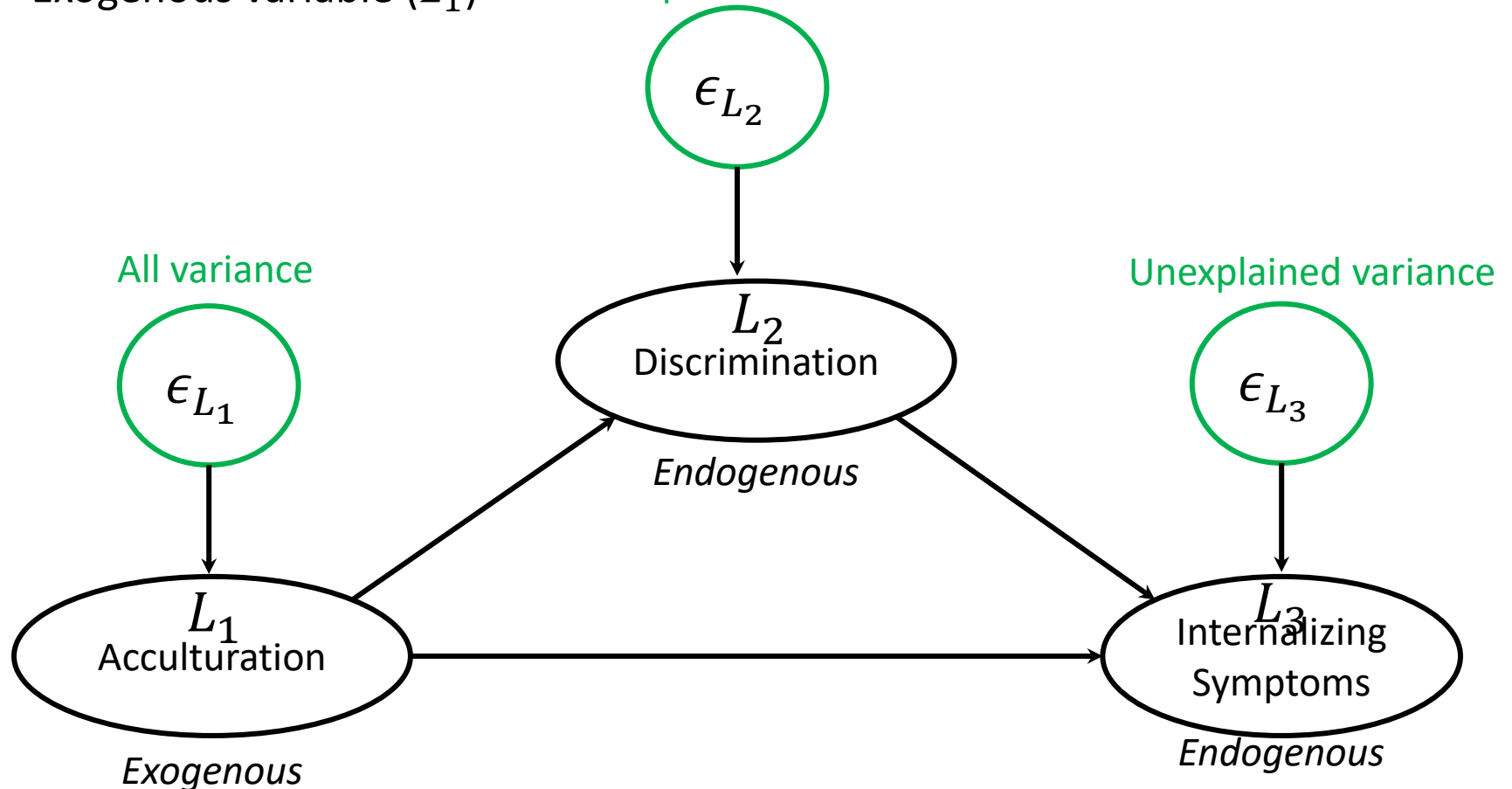
An Example: Internalizing among Latine Youth

Which variables need error terms?

Endogenous variables (L_2 and L_3)

Exogenous variable (L_1)

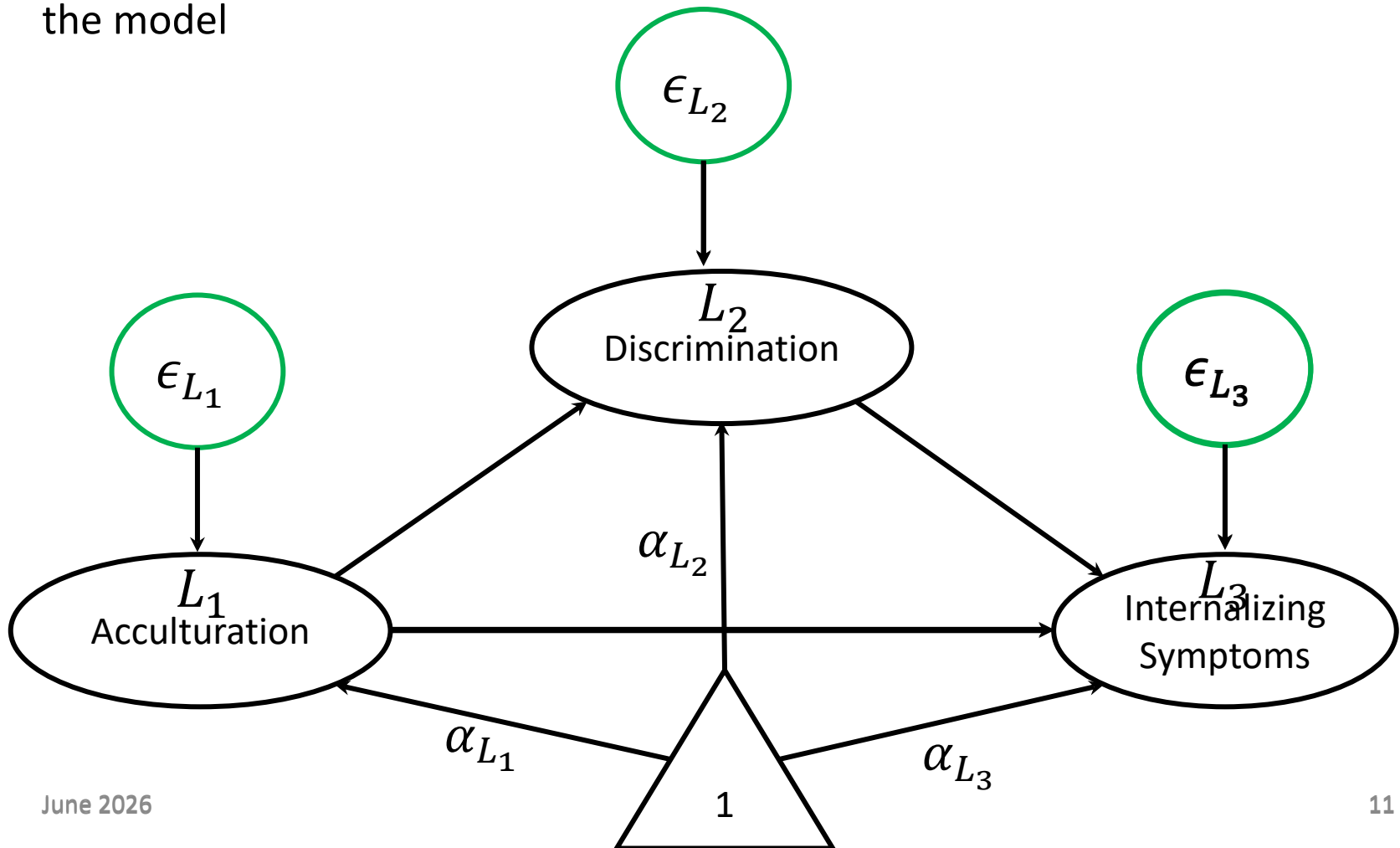
Unexplained variance



Intercepts

Each variable also needs an intercept, which represents the mean, or a general tendency shared across all cases.

Frequently omitted from path diagrams because not always relevant to the model



Latent Variable Model

Structural equations summarizing relationships among latent variables

Sometimes called “structural” model (but all equations are “structural”)

Consider our model:

Note: we'll come back to path diagrams in the next lecture.

Latent Variable Model:

$$L_{1i} = \alpha_{L_1} + \epsilon_{L_{1i}}$$

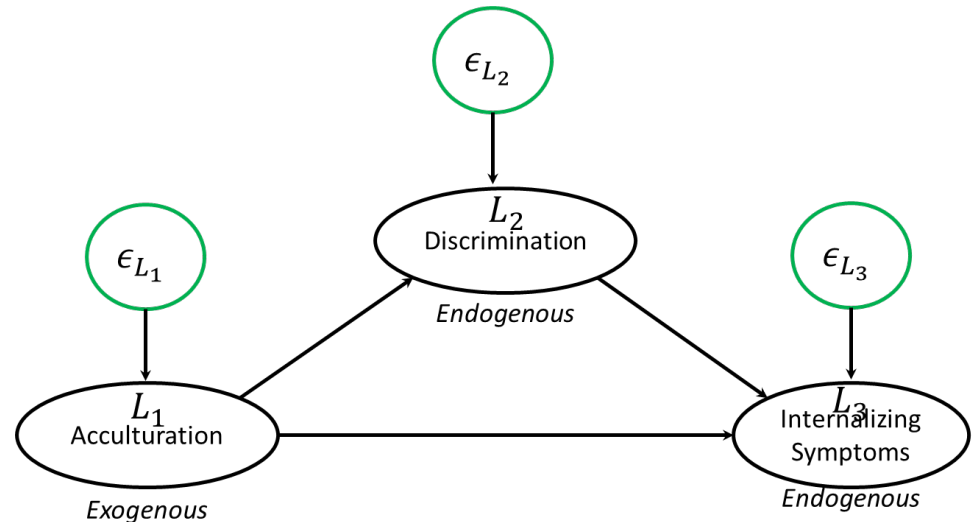
$$L_{2i} = \alpha_{L_2} + B_{21}L_{1i} + \epsilon_{L_{2i}}$$

$$L_{3i} = \alpha_{L_3} + B_{31}L_{1i} + B_{32}L_{2i} + \epsilon_{L_{3i}}$$

Notice:

Variables have i subscript, because they vary across cases

Parameters do not, assumed to be same across all cases



We're going to break this down bit by bit

Breaking Down Equation

$$L_{1i} = \alpha_{L_1} + \epsilon_{L_{1i}}$$

- L_1 is first *exogenous* latent variable (e.g., acculturation)
- Model suggests there are two parts to this latent variable:
 1. Part that is constant for all individuals
 - α_{L_1} is the intercept (or regression constant), mean
 2. Variance, deviation from the mean
 - ϵ_{L_1} deviation from expectation (the mean)
 - Captures all factors that effect L_1 that are not in the model (i.e., all factors that effect L_1)
 - No estimated coefficient (i.e., assume coefficient of 1)

Breaking Down Equation

$$L_{2i} = \alpha_{L_2} + B_{21}L_{1i} + \epsilon_{L_{2i}}$$

- L_2 is first *endogenous* latent variable (e.g., discrimination)
- Model suggests there are three parts to first endogenous latent variable:
 1. Part that is constant for all individuals
 - α_{L_2} is the intercept (or regression constant)
 - When L_2 and L_1 are both mean-centered, $\alpha_2 = \overline{L_2}$
 2. Part related to exogenous latent variable L_1 (e.g., acculturation)
 - B_{21} is like a regression coefficient
 - “For two cases that differ by 1 unit on L_1 , we expect the case 1 unit higher on L_1 to be B_{21} units higher on L_2 ”
 - To-from notation (To L_2 from L_1) = B_{21}
 3. Unexplained part
 - ϵ_{L_2} deviation from expectation
 - No estimated coefficient (i.e., assume coefficient of 1)

Breaking Down Equation

$$L_{3i} = \alpha_{L_3} + B_{31}L_{1i} + B_{32}L_{2i} + \epsilon_{L_{3i}}$$

- L_3 is second *endogenous* latent variable (e.g., internalizing symptoms)
- Model suggests there are four parts to second endogenous latent variable:
 1. Part that is constant for all cases: α_{L_3}
 2. Part related to exogenous latent variable L_1 (e.g., acculturation)
 - B_{31} is like a regression coefficient
 - “For two cases that differ by 1 unit on L_1 , we expect the case 1 unit higher on L_1 to be B_{31} units higher on L_3 , controlling for L_2 ”
 - To-from notation (To L_3 from L_1) = B_{31}
 3. Part related to first endogenous variable L_2 (e.g., discrimination)
 - B_{32} is like a regression coefficient
 - “For two cases that differ by 1 unit on L_2 , we expect the case 1 unit higher on L_2 to be B_{32} units higher on L_3 , controlling for L_1 ”
 - To-from notation (to L_3 from L_2) = B_{32}
 4. Unexplained part
 - $\epsilon_{L_{3i}}$ deviation from expectation
 - No estimated coefficient (i.e., assume coefficient of 1)

Matrix Representation

For our specific case:

$$\begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} = \begin{bmatrix} \alpha_{L_1} \\ \alpha_{L_2} \\ \alpha_{L_3} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ B_{21} & 0 & 0 \\ B_{31} & B_{32} & 0 \end{bmatrix} \begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} + \begin{bmatrix} \epsilon_{L_1} \\ \epsilon_{L_2} \\ \epsilon_{L_3} \end{bmatrix}$$

General Matrix Representation: $\mathbf{L} = \boldsymbol{\alpha}_L + \mathbf{B}\mathbf{L} + \boldsymbol{\epsilon}_L$

$\mathbf{L}_{N_L \times 1}$ latent endogenous variables

N_L number of latent variables

$\boldsymbol{\alpha}_{L_{N_L \times 1}}$ intercept vector

one for each L

$\boldsymbol{\epsilon}_{L_{N_L \times 1}}$ disturbances vector

one for each L

$\mathbf{B}_{N_L \times N_L}$ coefficients for $L \rightarrow L$ paths

elements are B_{ij}

main diagonal is always $\mathbf{0}$

Assumptions

The expected values of the deviations are zero: $E(\epsilon_L) = \mathbf{0}$

ϵ_{Lx} are uncorrelated with ϵ_{Ly}

ϵ_L is assumed to be homoscedastic and **not autocorrelated**

Not Autocorrelated: SEM version of “independence” assumption

ϵ_{Lli} uncorrelated with ϵ_{Llj} for all i and j where $i \neq j$ for all l

i.e., $C(\epsilon_{Lli}, \epsilon_{Llj}) = 0$ for $i \neq j$ and for all l

Note: Deviations from separate equations can be correlated

$(I - B)$ is nonsingular (i.e., $(I - B)^{-1}$ exists)

This will allow for a **reduced form** equation

$$IL = \alpha_L + BL + \epsilon_L$$

$$IL - BL = \alpha_L + \epsilon_L \quad \text{by commutative property}$$

$$(I - B)L = \alpha_L + \epsilon_L \quad \text{by distributive property}$$

$$L = (I - B)^{-1}(\alpha_L + \epsilon_L) \quad \text{by multiplying by inverse (must exist)}$$

Covariance Matrices

$\Sigma_{\epsilon_L \epsilon_L}$ covariance matrix of errors of latent variables (ϵ_L)

elements are $\sigma_{\epsilon_{L_i} \epsilon_{L_j}}$

symmetric matrix (covariance matrix)

if variances are all 1, then it is a correlation matrix

Can be expressed as a partition of covariances of residuals of exogenous latent variables ($\Sigma_{\epsilon_{L_X} \epsilon_{L_X}}$) and covariances of residuals of endogenous latent variables ($\Sigma_{\epsilon_{L_Y} \epsilon_{L_Y}}$)

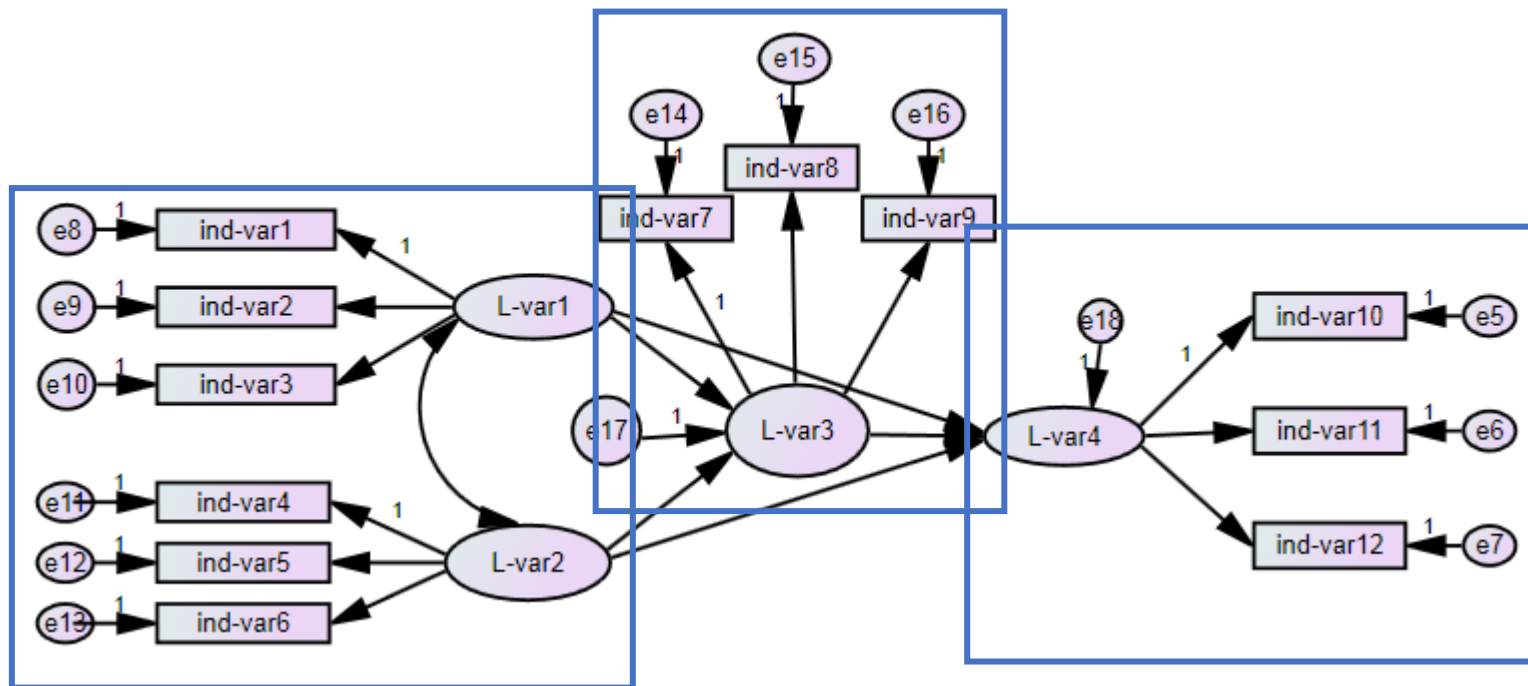
$$\Sigma_{\epsilon_L \epsilon_L} = \begin{bmatrix} \Sigma_{\epsilon_{L_X} \epsilon_{L_X}} & \mathbf{0} \\ \mathbf{0} & \Sigma_{\epsilon_{L_Y} \epsilon_{L_Y}} \end{bmatrix}$$

Zeros in off diagonal of partition come from assumption ϵ_{L_X} are uncorrelated with ϵ_{L_Y}

Subsystems of an SEM

There are two major parts of an SEM

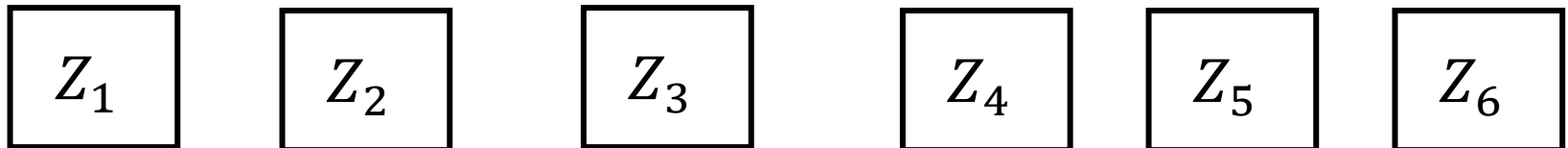
1. **Latent Variable Model: relations among latent variables**
2. **Measurement Model: relations among observed and latent variables**



Observed Variables

- Variables for which we actually have data (columns in our dataset)
 - Internalizing symptoms measured with 31 items from Youth Self Report
 - Acculturation measured with 12-item Short Acculturation Scale for Hispanic Youth
 - Discrimination measured with 10 items from Social, Attitudinal, Familial, and Environmental Acculturative Stress-Child Scale
- AKA: manifest variables, measures, indicators, *proxies*
- For now we'll assume these are *approximately continuous*

More on this later, but represented as squares/rectangles in path diagram



Measurement Model

Structural equations linking observed variables (Z_i) to latent variables (L_i)

Sometimes it is helpful to split up $Z = \begin{bmatrix} X \\ Y \end{bmatrix}$

X are **indicators** of exogenous latent variables

Y are **indicators** of endogenous latent variables

You can also remember because in regression X is predictor (exogenous) and Y is outcome (endogenous)

$$Z_i = \alpha_Z + \Lambda L_i + \epsilon_{Z_i}$$

Example model with 3 Z s indicating 1 L :

$$\begin{aligned} Z_{1i} &= \alpha_{Z_1} + \lambda_{11}L_{1i} + \epsilon_{Z_{1i}} \\ Z_{2i} &= \alpha_{Z_2} + \lambda_{21}L_{1i} + \epsilon_{Z_{2i}} \\ Z_{3i} &= \alpha_{Z_3} + \lambda_{31}L_{1i} + \epsilon_{Z_{3i}} \end{aligned}$$

$$\Lambda_{3 \times 1} = \begin{bmatrix} \lambda_{11} \\ \lambda_{21} \\ \lambda_{31} \end{bmatrix}$$

The Bits and Pieces

$$Z_i = \alpha_Z + \Lambda L_i + \epsilon_{Z_i}$$

λ s are like regression coefficients

“For two people who differ by one unit on L_j we expect the person one unit higher on L_j to be λ_{kj} units higher on Z_k ”

Factor variance is often set to 1, so 1 unit difference = 1 SD difference

Commonly called **factor loadings**

ϵ_{Z_i} are **errors of measurement** for their respective indicators

The part of the indicator that is not related to the latent variable

Sometimes called **unique factors** or **uniquenesses**

Assumptions

$$E(\epsilon_Z) = 0$$

Expected value of all ϵ s is zero

Uniquenesses are uncorrelated with L and Z s

$$C(\epsilon_{Z_{Xj}}, \epsilon_{Z_{Yk}}) = 0 \text{ for all } j, k$$

There is no covariance between uniquenesses between indicators of exogenous and endogenous variables

Uniquenesses are homoskedastic and not autocorrelated

Summary

Latent variable (L) are unobserved variables, often abstract or conceptual, which have no measurement error

Relationships between latent variables described using B matrix

Observed variables (Z) are assumed to be caused by latent variables (L)

Relationship between observed variables and latent variables are described using Λ matrix

Both observed and latent variables have intercepts (α) which are constant across all cases (parameter).

Both observed and latent variables have and errors/deviations (ϵ) which are assumed to vary across cases (random variable).

References

For more on expectations, pdfs, cdfs: Cassella & Berger Chpt 1 and 2

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